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Lysosomal enzyme trafficking factor LYSET enables nutritional usage of extracellular proteins

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Mammalian cells can generate amino acids through macropinocytosis and lysosomal breakdown of extracellular proteins, which is exploited by cancer cells to grow in nutrient-poor tumors. Here, through genetic screens in defined nutrient conditions we characterized LYSET, a transmembrane protein (TMEM251) selectively required when cells consume extracellular proteins. LYSET was found to associate in the Golgi with GlcNAc-1-phosphotransferase, which targets catabolic enzymes to lysosomes through mannose-6-phosphate modification. Without LYSET, GlcNAc-1-phosphotransferase was unstable owing to a hydrophilic transmembrane domain. Consequently, LYSET-deficient cells were depleted of lysosomal enzymes and impaired in turnover of macropinocytic and autophagic cargoes. Thus, LYSET represents a core component of the lysosomal enzyme trafficking pathway, underlies the pathomechanism for hereditary lysosomal storage disorders, and may represent a target to suppress metabolic adaptations in cancer.

When nutrients are abundant, mammalian cells preferentially meet their demands for exogenous amino acids by importing monomeric amino acids through plasma membrane transporters. However, most extracellular biomass in circulation and the extracellular space is contained within proteins, which can be broken down into amino acids intracellularly through endocytosis and lysosomal catabolism (1). By tapping into the copious nutrient stores of extracellular proteins, cells sustain a supply of bioenergetic and biosynthetic precursors during periods of starvation. Macropinocytosis, a non-selective endocytic pathway, and lysosomal catabolic activity are up-regulated by multiple oncogenic signaling pathways and frequently elevated in cancer (2, 3). The resulting increase in uptake and lysosomal breakdown of extracellular proteins constitutes an alternative amino acid source that enables malignant cells to grow in nutrient-poor tumor environments (4–6).

The nutrient environment profoundly influences the activities and traits of mammalian cells (7). Genetic screens have been instrumental in functionally annotating mammalian genes and identifying context-specific dependencies in cancer cell lines, but such screens are commonly performed under unphysiological nutrient conditions. For example, the balance between monomeric and protein-contained essential amino acids in standard cell culture media is skewed by 1–2 orders of magnitude compared to human plasma (8, 9). As a corollary, the molecular pathways that support a metabolic state in which cells survive and grow by using extracellular proteins as nutrients remain incompletely understood. Here, through proliferation-based CRISPR screens in defined nutrient conditions, we identify the transmembrane protein LYSET as a core component of the lysosomal enzyme trafficking pathway that is critical for lysosomal physiology, nutritional usage of extracellular proteins, and metabolic adaptations in cancer.

Results

A CRISPR screen identifies LYSET (TMEM251) as selectively required when cancer cells feed on extracellular proteins

To identify cellular pathways that are required for nutrient generation from extracellular proteins, we established screening conditions under which cell growth was supported either by uptake of monomeric amino acids or by macropinocytosis and lysosomal catabolism of extracellular proteins (Fig. 1A). To selectively enforce either nutrient acquisition pathway, the essential amino acid leucine was provided in its monomeric form or contained within albumin, the major human plasma protein (Fig. 1B). As primary screening model, we chose MIA PaCa-2 cells, which are derived from pancreatic adenocarcinoma, a nutrient-deficient tumor type where extracellular proteins constitute an important amino acid source (4–6). Indeed, when albumin was supplemented at physiological levels (3 – 4%), MIA PaCa-2 cells could sustain proliferation in leucine-deficient media through lysosomal albumin catabolism (fig. S1A).

To initiate gene editing at the onset of nutrient deprivation, we engineered a single cell-derived MIA PaCa-2 clone harboring doxycycline-inducible Cas9 (iCas9; fig. S1B) (10). This cell line did not display basal Cas9 activity, rapidly triggered CRISPR editing upon doxycycline addition, and robustly proliferated using albumin as a leucine source (fig. S1, C to E). MIA PaCa-2 iCas9 cells were transduced with the Vienna genome-wide sgRNA library (11) and, upon Cas9 induction, passaged in four media conditions that were either rich or deficient in leucine and/or albumin (Fig. 1C and fig. S1F). By quantifying changes in sgRNA representation in the different culture conditions over time, we identified multiple genes that were selectively required when albumin was an obligatory nutrient (data S1 and S2). These included endolysosomal trafficking regulators, amino acid transporters, and components of the amino acid-sensing mTORC1 and GCN2-ATF4 pathways (Fig. 1D and fig. S1G), which were validated to be selectively essential when albumin was an obligatory leucine source (fig. S2A).

One prominent hit that was selectively essential when cells fed on albumin was the transmembrane protein TMEM251 (Fig. 1D and fig. S1G), which hereafter is referred to as Lysosomal Enzyme Trafficking Factor (LYSET). To validate this selective dependency, we generated LYSET inducible knockout (iKO) MIA PaCa-2 cells (fig. S2B), as well as single cell-derived LYSET knockout (KO) clones from parental MIA PaCa-2 cells (Fig. 1E). LYSET KO cells grew normally under standard culture conditions. However, LYSET KO cells failed to proliferate when placed in leucine-deficient, albumin-supplemented medium (Fig. 1F) and were rapidly outcompeted in competitive proliferation assays (fig. S2C). We deleted LYSET in a range of human and murine cell types,

including additional pancreatic cancer cell lines, colon, lung and bladder cancer cells, as well as non-transformed mouse embryonic fibroblasts (MEFs) and immortalized baby mouse kidney (iBMK) cells with/without transformation by Hras^{G12V} to investigate LYSETs function more broadly. In line with CRISPR screens from the DepMap project, which characterize LYSET (TMEM251) as a widely non-essential gene (12), loss of LYSET had no effect on cell proliferation under standard culture conditions in any of the examined cell lines (Fig. 1G and fig. S3A). By contrast, in leucine-deficient medium supplemented with 3% albumin, loss of LYSET strongly suppressed proliferation and viability of all cancer cell lines, MEFs and iBMK cells. Hras^{G12V} transformation, which promotes macropinocytic nutrient uptake (4), enabled iBMK cells to grow by feeding on albumin, which was potentially suppressed by genetic ablation of LYSET (Fig. 1G).

To determine the relevance of LYSET for cancer cell growth in vivo, we examined the consequences of LYSET deficiency in several tumor models. Genetic ablation of LYSET impaired tumor growth of human MIA PaCa-2 and murine EPP2 pancreatic cancer cells upon subcutaneous transplantation into immunodeficient mice (fig. S3, B and C). To probe these effects in the presence of a functional immune system, we examined pro-tumorigenic functions of LYSET in a syngeneic colon cancer model (MC-38) and in an orthotopic transplantation model of pancreatic cancer (KPC). LYSET-deficient MC-38 cells were strongly impaired in subcutaneous tumor growth (Fig. 1, H and I), and LYSET-deficient KPC cells were rapidly outcompeted by control cells in orthotopic pancreatic tumors (Fig. 1J), both in syngeneic immunocompetent and immunodeficient recipient mice. Thus, LYSET is dispensable for cell viability and growth in nutrient-rich conditions but selectively required when extracellular proteins are obligatory nutrients, in amino acid-depleted conditions in vitro and during tumor growth in vivo.

LYSET is required for lysosomal degradation of macropinocytic and autophagic cargoes

To determine the subcellular process during which LYSET is required for the nutritional use of extracellular proteins, we explored possible functions in macropinocytosis, endosomal trafficking, and lysosomal catabolism. Albumin uptake and endosomal cargo trafficking to lysosomes were not altered in the absence of LYSET (fig. S4, A and B). However, lysosomal catabolism of a fluorescently labeled bovine serum albumin that becomes dequenched upon degradation (DQ BSA) was strongly decreased (Fig. 2, A and B). Lysosomal catabolism is a point of convergence of two metabolic pathways – macropinocytosis and autophagy, which supply extracellular and intracellular constituents, respectively (1, 3). Loss of LYSET led to strong accumulation of the autophagosomal proteins p62 and LC3-II, similar to pharmacological

suppression of lysosomal catabolism by the V-ATPase inhibitor bafilomycin A1 (Fig. 2C). Lysosomes break down biological macromolecules through the action of various hydrolytic enzymes. In LYSET-deficient cells, enzymatic activities of the lysosomal proteases cathepsins B and L were strongly reduced, comparable to the effect of pharmacological protease inhibition (Fig. 2D). Similarly, activities of the lysosomal glycosidases β -galactosidase and α -mannosidase were barely detectable. Thus, the failure to degrade macropinocytic and autophagic cargoes in the absence of LYSET is caused by a loss of lysosomal enzyme activity. Consequently, albumin-supported cell proliferation was suppressed to a comparable extent by LYSET deletion and by lysosomal protease inhibition (fig. S4C). In vivo, unlike LYSET deficiency, genetic ablation of the essential autophagy component ATG5 did not significantly impair tumor growth of MC-38 cells (fig. S4, D and E). Thus, lysosomal catabolism is required for tumor formation in this model, whereas autophagic turnover of intracellular constituents is largely dispensable.

LYSET is a core component of the mannose-6-phosphate pathway for lysosomal enzyme trafficking

To understand why LYSET was required for the catabolic activity of lysosomes, we determined changes in the proteome of LYSET KO MIA PaCa-2 cells using liquid chromatography-mass spectrometry and label-free quantification. From ~2,800 robustly quantified proteins, < 6% were significantly changed in LYSET-deficient cells (adj. p-value < 0.05). Autophagic cargo receptors were among the most enriched proteins, consistent with the block in autophagosomal protein turnover (Fig. 3A and data S3). We noted that LYSET-deficient cells displayed a global decrease in lysosomal enzymes and other lysosomal luminal proteins. To more precisely determine changes in the lysosomal proteome, we enriched iron nanoparticle-loaded lysosomes over magnetic columns. From 49 quantified lysosomal luminal proteins, almost all were strongly depleted in LYSET-deficient cells (Fig. 3B and data S3). By contrast, the abundance of integral and peripheral lysosomal membrane proteins was not altered in cell lysates (fig. S5A) and lysosomal fractions (fig. S5B). This suggests that LYSET is dispensable for lysosomal biogenesis per se, but specifically required for lysosomal accumulation of catabolic enzymes and other luminal proteins.

Most lysosomal enzymes are synthesized as immature pro-enzymes and subsequently activated through proteolytic processing when reaching the lysosome (13). LYSET-deficient MIA PaCa-2 cells did not display decreased mRNA levels of cathepsin B, cathepsin L and β -hexosaminidase (fig. S5C) or decreased abundance of their pro-enzyme forms (Fig. 3C). By contrast, the mature enzymes were nearly undetectable (Fig. 3C). Similarly, loss of LYSET led to a strong decrease in mature lysosomal enzymes in other non-transformed and

cancer-derived human and murine cell types (Fig. 3D and fig. S5, D and E). To trace the fate of the lysosomal enzymes, we determined their levels in the extracellular space. Immature pro-forms of cathepsins B and L were substantially increased in supernatants from LYSET-deficient cells (Fig. 3E). Consistently, secretome analysis revealed that lysosomal luminal proteins were generally enriched in cellular supernatants (fig. S5F and data S3). Thus, without LYSET most proteins normally destined for the lysosomal lumen were mistrafficked to the extracellular space.

Upon reaching the Golgi, most newly synthesized lysosomal enzymes receive mannose-6-phosphate (M6P) modifications on N-linked glycan chains through consecutive action of GlcNAc-1-phosphotransferase (GNPT) and M6P-uncovering enzyme (UCE), which serve as a lysosomal trafficking signal for M6P receptors (Fig. 3F) (13, 14). To examine the status of M6P modifications in LYSET-deficient cells, we used a single-chain antibody fragment that specifically detects M6P-containing proteins (15). Loss of LYSET caused a substantial decrease in cellular M6P-modified proteins (Fig. 3G and fig. S6A). When secretion of newly synthesized lysosomal enzymes was induced by ammonium chloride, M6P-modified proteins were readily detected in supernatants from control cells but not LYSET-deficient cells (Fig. 3H and fig. S6A). This suggested that LYSET functions as a component of the M6P pathway. Consistently, GNPTAB, which encodes the α and β subunits of the GlcNAc-1-phosphotransferase complex (16, 17), scored in our CRISPR screen as selectively required for albumin-supported growth (Fig. 1D).

To further investigate the relationship between LYSET and GNPTAB, we compared the phenotypic consequences of deleting either gene. In multiple cell lines, genetic ablation of LYSET or GNPTAB caused a comparable, strong reduction in M6P modifications (Fig. 3, G and H) and mature lysosomal enzymes (Fig. 3G and fig. S6, B and C), as well as increased secretion of immature pro-enzymes (fig. S6D). Secretome analysis from LYSET- and GNPTAB-deficient MEFs demonstrated indistinguishable hypersecretion of lysosomal enzymes (Fig. 3I and data S4). Consistently, a deficiency in LYSET or GNPTAB had the same metabolic consequences: accumulation of p62 and LC3-II (fig. S6, B and C), loss of lysosomal catabolic activity (fig. S6, E to H), and suppression of cell proliferation with extracellular proteins as nutrients (fig. S6I). Besides GNPTAB, two other proteins participate in the generation of M6P groups: GNPTG, which encodes the γ subunit of GlcNAc-1-phosphotransferase, and UCE, which cleaves the terminal GlcNAc residue from GlcNAc-phosphomannose groups (13, 14). Loss of GNPTG or UCE did not cause an overt reduction in mature lysosomal enzymes (fig. S6C), lysosomal proteolysis (fig. S6H), or extracellular protein-supported cell proliferation (fig. S6I). This is consistent with the accessory function of GNPTG (18), and the ability of M6P

receptors to recognize both products and substrates of UCE (19).

Because the lysosomal enzyme trafficking pathway originates in the Golgi, we wondered whether LYSET generally affected Golgi function. LYSET-deficient cells did not display apparent changes in Golgi morphology (fig. S7, A and B) or the Golgi proteome (Fig. 3A). LYSET-deficient cells were also not impaired in constitutive protein secretion (Fig. 3I) or lysosomal transport of lysosomal acid glucosylceramidase (GBA), which uses an M6P-independent trafficking pathway (Fig. 3B and fig. S5F). Thus, LYSET is an essential and specific component of the M6P pathway for lysosomal enzyme trafficking.

LYSET deficiency causes lysosomal storage disorder phenotypes

Hereditary mutations in GNPTAB that lead to a loss or reduction in GlcNAc-1-phosphotransferase activity cause mucopolidosis type II and III, respectively, autosomal recessive lysosomal storage disorders characterized by varying degrees of skeletal abnormalities, cardiomegaly, and developmental delay (20). Because LYSET KO cells were deficient in M6P modifications, we wondered whether they displayed similar lysosomal storage disorder phenotypes. Electron microscopy of LYSET-deficient MEFs and MIA PaCa-2 cells revealed that lysosomes often contained electron-dense material in their lumen, indicative of impaired digestion of lysosomal contents (Fig. 4A). Moreover, lysosomes were often enlarged, more irregularly sized, and accumulated to high numbers in the cytosol (Fig. 4, B to D, and fig. S7C). The accumulation of lysosomes in LYSET-deficient cells was also apparent by the strong increase in lysotracker-positive organelles (fig. S7, D to F). These lysosomal defects resembled the phenotypes of GNPTAB-deficient cells (Fig. 4, A to D, and fig. S7, C to F), as well as the phenotypes of fibroblasts from mucopolidosis II/III patients (20, 21).

Recently, two LYSET variants, R45W and Y72X, have been linked to familial skeletal dysplasia syndromes reminiscent of mucopolidosis II/III (22). To determine the functional impact of these disease-associated LYSET mutations, we stably expressed LYSET^{R45W}, LYSET^{Y72X}, or wild type LYSET in HEK 293T and MIA PaCa-2 cells in which endogenous LYSET was deleted. In contrast to wild type LYSET, which fully restored mature cathepsin levels and suppressed LC3-II accumulation (fig. S8A), LYSET^{R45W} and LYSET^{Y72X} failed to rescue cathepsin and LC3-II levels (Fig. 4E and fig. S8, B and C). Consistently, LYSET^{R45W} and LYSET^{Y72X}-expressing cells displayed decreased lysosomal enzyme activities (Fig. 4F) and accumulated lysotracker-positive organelles (Fig. 4, G and H). LYSET^{R45W} and LYSET^{Y72X} were barely detectable at the protein level, conceivably because they abrogated LYSET function by destabilization and premature truncation of the

protein, respectively (Fig. 4E and fig. S8, B and C). Phenotypically, this resembled disease-associated mutations in GNPTAB that destabilize the protein (fig. S8, C and D). Thus, loss-of-function mutations in LYSET lead to lysosomal storage disorder phenotypes at the cellular level that are characteristic of mucopolidosis II/III.

Retention of GlcNAc-1-phosphotransferase in Golgi membranes depends on association with LYSET

To understand the molecular function of LYSET in the generation of M6P groups, we investigated possible interactions with GNPTAB. LYSET was present in the Golgi (fig. S9A), where it co-localized with epitope-tagged GNPTAB (Fig. 5A). Proximity ligation yielded a strong signal between GNPTAB and LYSET (Fig. 5, B and C), but not for either of the two proteins with other ubiquitous Golgi proteins (fig. S9B). AlphaFold modeling predicted a high-confidence interaction between LYSET and GNPTAB (fig. S9, C to F), which was validated by efficient, reciprocal co-immunoprecipitation of transiently overexpressed LYSET and GNPTAB (Fig. 5, D and E), as well as endogenous LYSET and ectopically expressed GNPTAB (fig. S9, G and H). SILAC-CoIP proteomics of the endogenous proteins revealed that LYSET specifically interacted with GNPTAB, GNPTG, and several GlcNAc-1-phosphotransferase substrates (Fig. 5F and data S5). These results establish LYSET as a component of the Golgi-resident GlcNAc-1-phosphotransferase complex.

Next, we examined a potential role of LYSET in controlling the abundance of mature GNPTAB. Upon reaching the Golgi, the GNPTAB α/β precursor is cleaved into α and β subunits by site-1 protease (S1P) (23). We monitored mature GNPTAB by using an antibody that recognizes the endogenous α subunit and by ectopically expressing GNPTAB-myc, which generates an epitope-tagged β subunit. LYSET-deficient cells did not display a decrease in GNPTAB mRNA (fig. S9I) or the α/β precursor protein (Fig. 5G). However, LYSET-deficient cells displayed an almost complete loss of the mature GNPTAB α and β subunits (Fig. 5, G and H, and fig. S9J). β subunit levels were also decreased in LYSET^{R45W} cells, consistent with the poor expression of this mutant LYSET variant (fig. S9K). However, LYSET^{R45W} and GNPTAB still co-immunoprecipitated, suggesting that LYSET^{R45W} retains the interaction with GNPTAB (fig. S9, K and L). Conversely, GNPTAB-deficient cells did not display a decrease in LYSET expression (fig. S6, B and C) or Golgi localization (fig. S9M). Because these findings suggested that LYSET controls the abundance of GNPTAB at the posttranslational level, we examined GNPTAB processing. Radioactive pulse-chase experiments demonstrated that LYSET was dispensable for synthesis and proteolytic processing of ectopically expressed GNPTAB α/β precursor (fig. S10A). ATF6, another S1P substrate, was also cleaved normally (fig. S10B). Moreover,

endoglycosidase H_r (Endo H_r) and N-glycosidase F (PNGase F) treatments indicated that high mannose- and complex-type glycan modifications of mature GNPTAB were not perturbed in the absence of LYSET (fig. S10C).

Because GNPTAB processing appeared to be intact in LYSET-deficient cells, we reasoned that LYSET was required for stabilization of the mature protein. To test this, we sought to block GNPTAB degradation by inhibiting lysosomal proteases or the proteasome, two pathways that degrade Golgi-derived proteins (24). In LYSET wild-type cells, bafilomycin A1 and lysosomal protease inhibitors led to a slight increase in the GNPTAB β subunit (fig. S10D). In LYSET-deficient cells, bafilomycin A1 stabilized the GNPTAB α and β subunits (Fig. 5, H and I) and restored GNPTAB levels in the Golgi (fig. S10E), to a similar degree as previously observed in cells expressing Golgi retention-deficient GNPTAB mutants (25). SIP inhibition abrogated accumulation of the GNPTAB α and β subunits in LYSET-deficient cells treated with bafilomycin A1, corroborating that cleavage of the α/β precursor by SIP occurs in this context (Fig. 5, H and I). Lysosomal protease inhibitors also stabilized the α and β subunits (Fig. 5, J and K), whereas proteasome inhibition did not have any effect (fig. S10, F and G). Thus, in the absence of LYSET, GNPTAB is lost by trafficking to lysosomes and subsequent degradation through residual protease activity.

To understand how LYSET facilitates proper localization of GNPTAB, we examined the transmembrane domain of GNPTAB, which consists of one N-terminal helix in the α subunit (TM1) and one C-terminal helix in the β subunit (TM2). TM1 of human GNPTAB contains multiple charged/hydrophilic residues, whose biophysical properties destabilize helices in a hydrophobic membrane environment (Fig. 6, A and B). By contrast, TM1 is a regular hydrophobic transmembrane helix in the GNPTAB homolog of *Drosophila*, which does not possess LYSET. Intriguingly, the unfavorable transmembrane domain of GNPTAB co-occurs evolutionarily with LYSET: While GNPTAB homologs in animals that lack LYSET contain hydrophobic transmembrane helices, GNPTAB homologs in animals that possess LYSET are predicted to insert poorly into membranes (Fig. 6C and data S6). At the sequence level, the presence of LYSET coincides with a conserved charged/hydrophilic patch in TM1 of GNPTAB (Fig. 6D). To test the functional relevance of this feature, we substituted two conserved charged/hydrophilic amino acid residues in TM1 of human GNPTAB (Q36 and E39; Fig. 6A) with leucine and investigated localization and function of the mutant protein in LYSET-deficient, wild type, and LYSET-overexpressing cells. As expected, wild type GNPTAB was undetectable in LYSET-deficient cells, while overexpression of LYSET stabilized GNPTAB at elevated levels (Fig. 6E). By contrast, GNPTAB^{Q36L,E39L} was highly expressed and localized to the Golgi independently of LYSET (Fig. 6, E and F, and fig. S10H).

Consequently, GNPTAB^{Q36L,E39L} fully rescued lysosomal protein degradation and proliferation of cells that utilize extracellular proteins as nutrients in LYSET-deficient cells (Fig. 6, G and H). Thus, replacing the unusual charged/hydrophilic motif of TM1 with hydrophobic residues restores GNPTAB function in the absence of LYSET.

Discussion

The above results identify LYSET as a core component of the M6P pathway that is required for lysosomal enzyme trafficking and, consequently, for lysosomal nutrient generation (fig. S11). LYSET is a small transmembrane protein of 131 amino acid residues (conserved isoform) that lacks discernible homology to other proteins. A clue for the molecular function of LYSET came from the unusual transmembrane domain of its interaction partner GNPTAB, which contains multiple charged/hydrophilic amino acid residues that energetically disfavor membrane integration. Structural modeling and biochemical analyses indicate that association of LYSET with the GNPTAB α and β subunits provides the free energy to form a stable transmembrane domain in the GlcNAc-1-phosphotransferase complex. LYSET is highly conserved in vertebrates and clear homologs are present in several other metazoan phyla. The co-occurrence of LYSET with GNPTAB homologs that possess an energetically disfavored transmembrane domain suggests that they became functionally linked during the evolution of the M6P pathway. Conceivably, controlling GlcNAc-1-phosphotransferase levels through LYSET constitutes a mechanism to regulate the lysosomal enzyme trafficking pathway.

A deficiency in either LYSET or GNPTAB caused indistinguishable cellular phenotypes: accumulation of lysosomes that are depleted of catabolic enzymes, fail to degrade macropinocytic and autophagic cargoes and contain undigested contents. Similar phenotypes were observed for mutant variants of LYSET that are associated with skeletal dysplasia syndromes which resemble the lysosomal storage disorders mucopolipidosis II/III (22). The present study provides a mechanistic explanation for disease-associated LYSET variants and suggests that loss-of-function mutations in LYSET could underlie mucopolipidosis II/III cases that lack mutations in other components of the GlcNAc-1-phosphotransferase complex. Two recent CRISPR screens identified LYSET as required for viral infections (accompanying paper by Richards *et al.*) and turnover of lysosomal membrane proteins (26). These findings further highlight the importance of LYSET in diverse physiological and pathological processes where lysosomal catabolism plays a critical role.

Our findings reveal an essential function of LYSET and the M6P pathway for cells that depend on lysosomal catabolism to generate nutrients. One context in which this metabolic state arises is in solid tumors, where limited or

dysfunctional vasculature creates nutrient-deficient regions. To cope with such austere environments, cancer cells commonly utilize lysosomal protein catabolism to acquire amino acids (1–3). Several oncogenic signaling pathways promote cellular processes that supply macromolecular substrates to lysosomal catabolism, thereby enhancing the metabolic flexibility and resilience of cancer cells. In particular, oncogenic Ras signaling induces macropinocytosis of extracellular proteins as exogenous nutrients (4), and elevates basal autophagy for recycling of intracellular constituents (27). Genetic ablation of LYSET potently suppresses lysosomal degradation of macropinocytic and autophagic cargoes but does not impair cell viability and growth under nutrient-replete conditions. Thus, strategies to inhibit LYSET and the lysosomal enzyme trafficking pathway may provide a promising entry point to therapeutically exploit metabolic adaptations in cancer.

Methods summary

Cell lines used in most experiments were MIA PaCa-2, HEK 293T and MEFs. Cell culture experiments in different nutrient conditions were performed with amino acid-free, glucose-free DMEM/F-12 reconstituted with glucose and all amino acids except leucine at standard DMEM/F12 concentrations. For MIA PaCa-2 cells, leucine-free medium was supplemented with 4% albumin, 10% FBS and 5 μ M leucine, which supports continuous proliferation. For other cell lines, leucine-free medium was supplemented with 3% albumin and 10% dialyzed FBS. In proliferation assays, cell numbers were determined using a CASY Cell Counter or a Guava easyCyte flow cytometer. In competitive proliferation assays, sgRNA-positive cells were co-cultured with non-sgRNA-expressing controls and percentages of sgRNA-positive cells quantified using a Guava easyCyte flow cytometer.

A MIA PaCa-2 iCas9 clone transduced with the genome-wide Vienna sgRNA library was described previously (11). Cas9 expression was induced with doxycycline for three days prior to the screen. On day 0, cells were plated in four media conditions that were either rich or deficient in monomeric leucine and/or albumin as two distinct leucine sources (AA-rich $\pm$ 4% albumin, Leu-deficient $\pm$ 4% albumin). Samples were collected on day 0, and after approximately 9 and 14 population doublings or after an equivalent period under selection in the different nutrient conditions. Cells corresponding to > 600-fold library representation were collected, next-generation sequencing libraries prepared, pooled and sequenced on an Illumina HiSeq 2500 platform. Depletion and enrichment of sgRNAs were calculated as described previously (11) using the MAGeCK algorithm (28) for each condition and time point by comparison to day 0. Time points of the same condition were merged, representing for each gene the maximal effect (more significant p-value). For the

conditions AA-rich and Leu-deficient + 4% albumin, changes in sgRNA abundance were also calculated by direct comparison of the end points.

For immunoblotting, endogenous and ectopically expressed GNPTAB were analyzed in organelle-enriched fractions. Cells were homogenized with a Dounce tissue grinder. From post-nuclear supernatants (PNS), organelle fractions were prepared by centrifugation at 18,000 g for 30 min or at 100,000 g for 60 min. Organelles were resuspended in lysis buffer and subjected to immunoblotting. For analysis of GNPTAB glycosylation, equal protein amounts of PNS were treated with the glycosidases Endo H_f or PNGase F for 3 hours. M6P-modified proteins were analyzed in organelle fractions and in supernatants upon induced secretion with 10 mM NH₄Cl using an M6P-scFv antibody (15).

Lysosomal DQ BSA fluorescence dequenching and lysotracker accumulation were quantified by live imaging or flow cytometry. Cells were incubated with DQ BSA for 6 hours, and with lysotracker for the final 1 hour. For live imaging, Hoechst was also added for the final 1 hour. Activities of α -mannosidase and β -galactosidase were assayed with the substrates 4-nitrophenyl- α -D-mannopyranoside and 4-nitrophenyl- β -D-galactopyranoside, respectively; absorbance of liberated p-nitrophenol was measured at 405 nm. Activities of cathepsins B and L were measured with the substrates Z-Arg-Arg-7-amido-4-methylcoumarin and Z-Phe-Arg-7-amido-4-methylcoumarin, respectively; fluorescence of liberated 7-amido-4-methylcoumarin was measured with excitation at 365 nm and emission at 440 nm.

For lysosomal fractionation, lysosomes were loaded with ferromagnetic nanoparticles (DexoMAG C) and enriched over LS MACS columns. For secretome analysis, cells were cultured in OptiMEM for 24 hours, media collected and cleared by centrifugation. For SILAC-based Co-IPs, LYSET wild type and KO cells were labeled with medium or heavy arginine and lysine isotopes for three passages; isotope labels were swapped between genotypes in different experimental replicates to avoid isotope bias. Cell lysates were pooled and immunoprecipitated using an antibody that recognizes endogenous LYSET. Sample preparation for proteomics was performed via tryptic in-gel digestion. Liquid chromatography-mass spectrometry analysis was performed on an Ultimate 3000 UPLC system connected to an Orbitrap Exploris 480 mass spectrometer using data-dependent acquisition (DDA) mode. Data analysis of LFQ and SILAC experiments was performed with MaxQuant (29) using an organism-specific database extracted from UniProt under default settings. The statistical analysis for LFQ values and SILAC ratios was performed with the R-package limma (30). p-values were adjusted with the Benjamini-Hochberg method for multiple testing.

For structural modeling, structures of human and *Drosophila* GNPTAB were predicted five times using a local version of ColabFold (31); transmembrane helix 1 of the model with the best pLDDT scores was used. To obtain a model of the transmembrane region of the complex between human LYSET and GNPTAB, the analysis was restricted to the immediate membrane region, which yielded high pLDDT and PAD scores, indicating reliable models. For ΔG_{app} prediction, GNPTAB orthologs were identified using NCBI blast in the UniProt reference proteomes database or the NCBI non-redundant protein database. LYSET/TMEM251 orthologs were collected in a search using the Pfam TMEM251 hidden Markov model in the NCBI non-redundant protein database (32). Species were grouped by the presence or absence of a detectable TMEM251 ortholog. GNPTAB transmembrane helices 1 and 2 were predicted with TMHMM (33). The apparent free energy difference, ΔG_{app} , for insertion of the transmembrane sequence into the endoplasmic reticulum membrane was predicted with ΔG predictor (34).

For subcutaneous tumor growth experiments, suspensions of LYSET KO and control cells (EPP2, MC-38, or MIA PaCa-2) were injected into shaved flanks of anesthetized mice. Tumor size was monitored by caliper measurements two to three times a week. Necropsies were taken when termination criteria were reached as defined by the animal protocol. For orthotopic competition assays, LYSET KO and control KPC cells were pooled in a ratio of 12.5:1 and injected into the pancreas of anesthetized mice. Animals were sacrificed 14 days after orthotopic injection, and the ratio of LYSET KO to control cells was quantified by FACS.

A full description of the materials and methods including primer and sgRNA sequences (tables S1 to S3) is provided in the supplementary materials.

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SUPPLEMENTARY MATERIALS

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Materials and Methods

Figs. S1 to S11

Tables S1 to S3

References (35–48)

MDAR Reproducibility Checklist

Data S1 to S6

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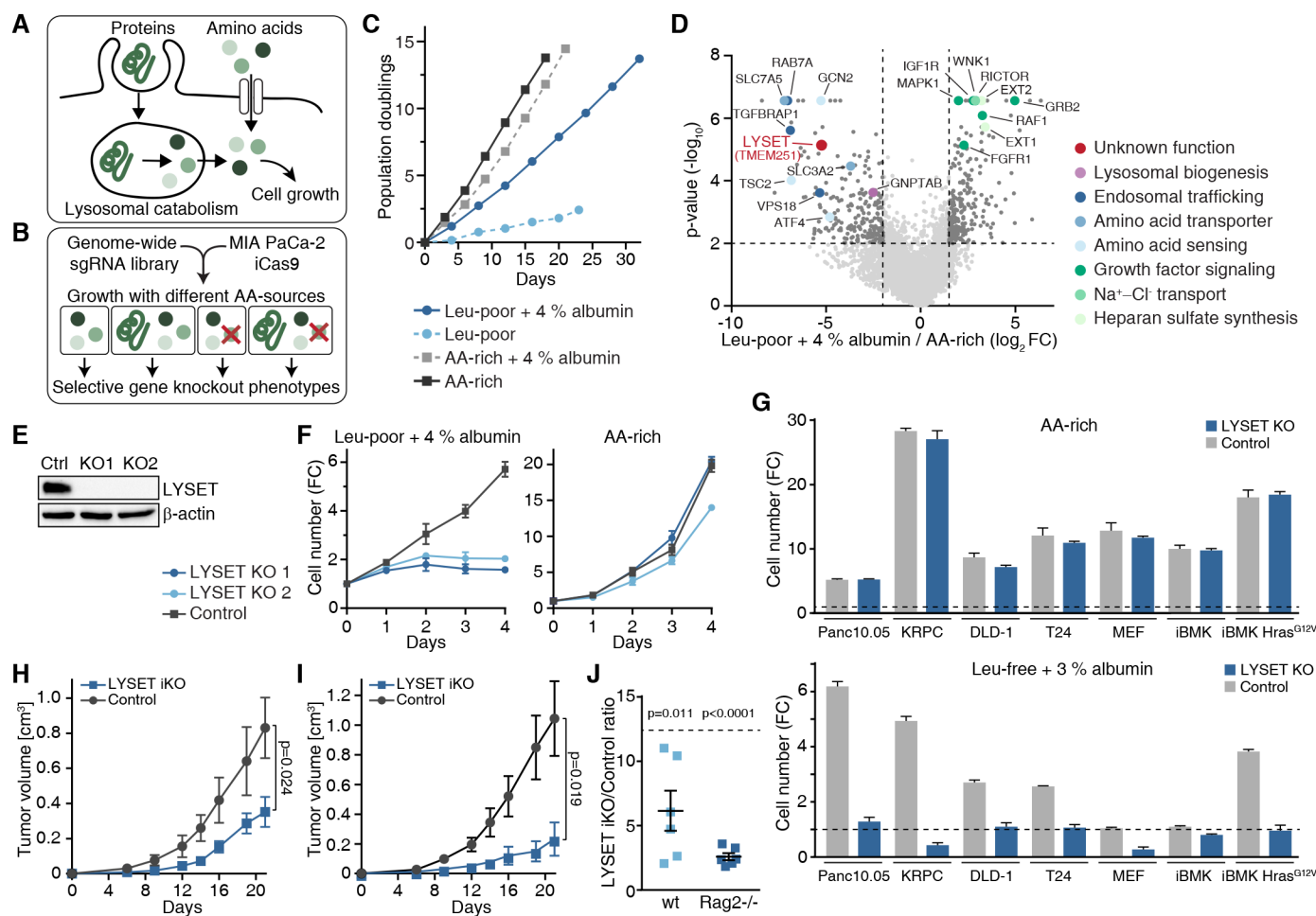


Fig. 1. A genome-wide CRISPR screen identifies LYSET as selectively essential when cancer cells utilize extracellular proteins as nutrients. (A) Mammalian cells can acquire exogenous amino acids (AA) through import of monomeric amino acids or through endocytosis and lysosomal catabolism of extracellular proteins. (B) Experimental design of the CRISPR screen to identify regulators of cell proliferation supported by extracellular proteins as nutrients. (C) Population doublings of MIA PaCa-2 cells during the CRISPR screen. (D) Gene-level enrichment or depletion of sgRNAs in leucine-deficient + 4% albumin medium versus AA-rich medium in the CRISPR screen. Dashed lines indicate significance ($p < 0.01$) of genes that are essential ($\log_2$ fold change (FC) < -2) or detrimental ($\log_2$ FC > 1.5) when cells feed on extracellular proteins. Selected hits are highlighted. (E) Immunoblot (IB) of LYSET KO MIA PaCa-2 cells. (F) Proliferation of LYSET KO MIA PaCa-2 cells in leucine-deficient + 4% albumin medium and in AA-rich medium. Data are represented as mean $\pm$ SD (N = 3). (G) Fold change in cell number of LYSET KO cell lines after 3 days in AA-rich medium or 4 days in leucine-deficient + 3% albumin medium. Data are represented as mean $\pm$ SD (N = 3). Dashed lines indicate starting cell numbers. (H and I) Growth of subcutaneous syngeneic tumors from LYSET iKO MC-38 cells in (H) immunodeficient Rag2^{-/-} mice (Control N = 6, LYSET iKO N = 7), (I) wild type mice (N = 5). Data are represented as mean $\pm$ SEM. (J) Competitive growth of orthotopic pancreatic tumors from LYSET iKO KPC cells in syngeneic wild type or Rag2^{-/-} mice. The LYSET iKO/control cell ratio after 14 days, quantified by flow cytometry, is represented as mean $\pm$ SEM (N = 6); the dashed line indicates the initial ratio. For the final time point of tumor growth experiments, p-values were calculated by unpaired two-sided *t* test with Welch correction.

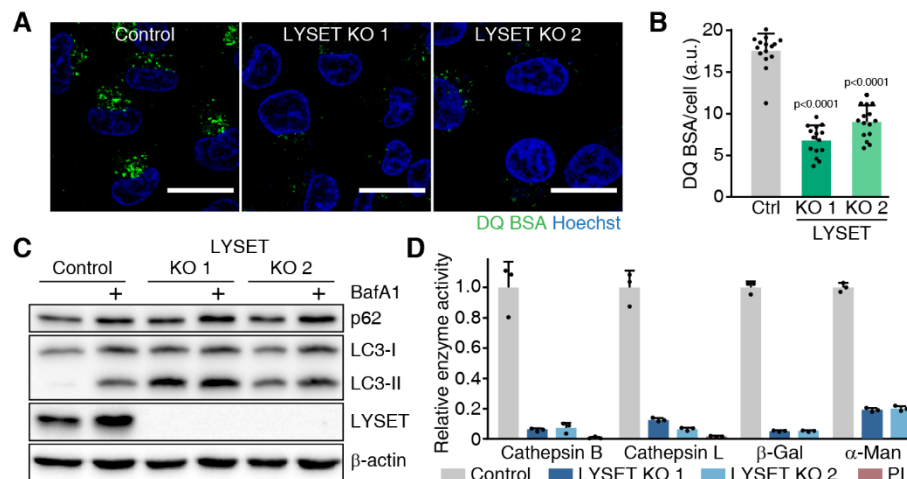


Fig. 2. LYSET is required for lysosomal catabolism of macropinocytic and autophagic cargoes. (A) Lysosomal DQ BSA degradation in LYSET KO cells, analyzed by microscopy. Scale bars = 20 μ m. (B) Quantification of DQ BSA degradation of cells shown in (A). Data are represented as mean $\pm$ SD (N = 15 fields of view with $\geq$ 10 cells). p-values were calculated by unpaired two-sided *t* test with Welch correction. (C) p62 and LC3 levels in LYSET KO cells $\pm$ bafilomycin A1 [100 nM] for 3 hours, analyzed by IB. (D) Enzymatic activity of cathepsin B, cathepsin L, β -galactosidase and α -mannosidase in LYSET KO cells. Cathepsins B and L activities were also measured in control cells + protease inhibitors (PI). Data are represented as mean $\pm$ SD (N = 3). Experiments were performed in MIA PaCa-2 cells.

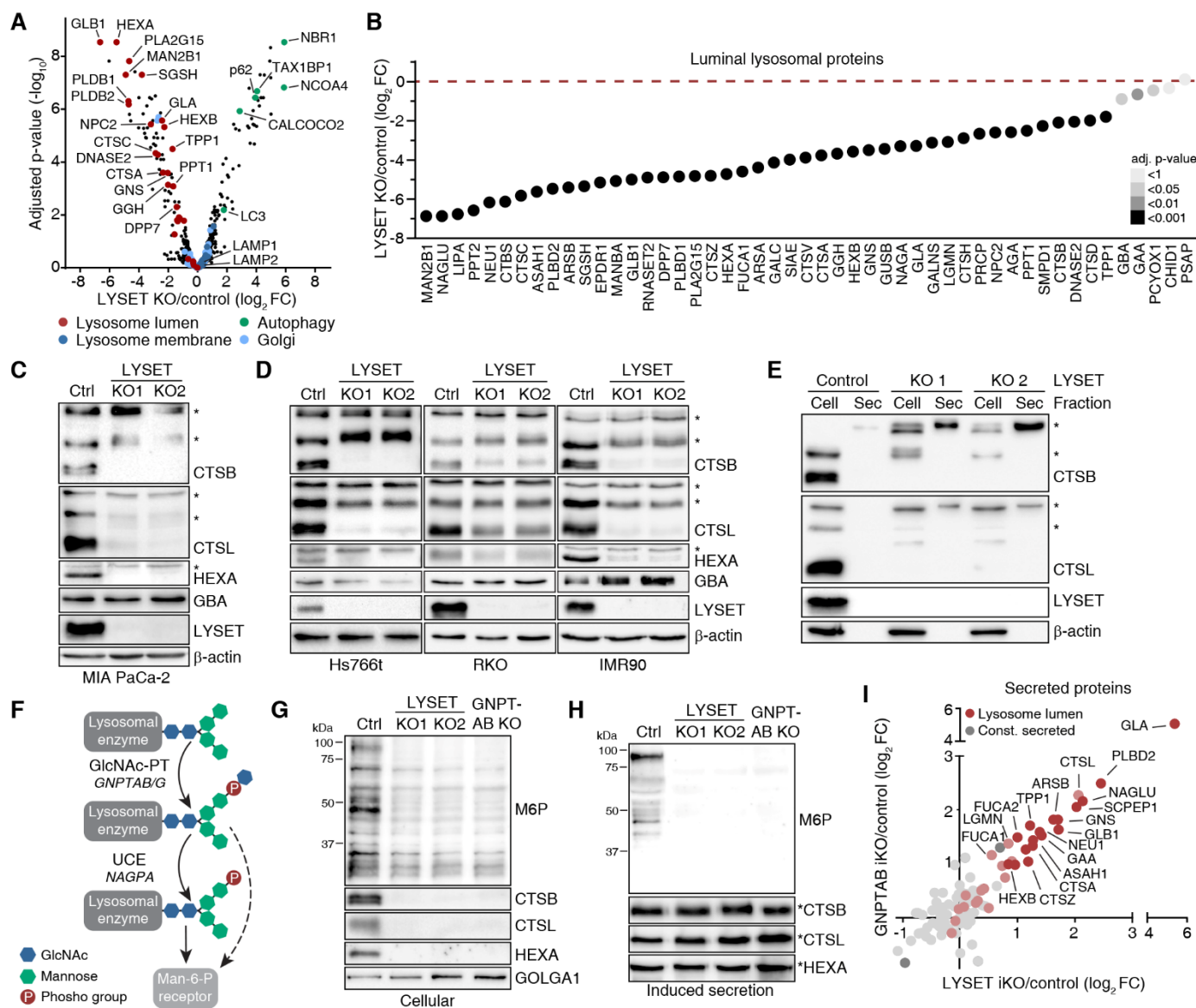


Fig. 3. LYSET is required for mannose-6-phosphate-dependent trafficking of lysosomal enzymes. (A) Changes in the proteome of LYSET KO MIA PaCa-2 cells, quantified by label-free mass spectrometry (N = 5). (B) Changes in lysosomal luminal proteins in enriched lysosomal fractions of LYSET KO MIA PaCa-2 cells, quantified by label-free mass spectrometry (N = 5). (C and D) Mature lysosomal enzymes in indicated LYSET KO cancer cell lines and IMR90 fibroblasts, analyzed by IB. (E) Secretion of immature cathepsins by LYSET KO MIA PaCa-2 cells, analyzed by IB. (F) Schematic of the M6P pathway. GlcNAc-PT: GlcNAc-1-phosphotransferase (the α and β subunits are encoded by *GNPTAB*, the γ subunit by *GNPTG*); UCE: uncovering enzyme (encoded by *NAGPA*). (G and H) M6P modifications in (G) organelle fractions, (H) supernatants upon NH_4Cl -induced secretion in LYSET KO and *GNPTAB* KO MIA PaCa-2 cells, analyzed by IB. (I) Changes in luminal lysosomal proteins and constitutively (const.) secreted proteins in the secretome of LYSET KO and *GNPTAB* KO MEFs, quantified in cellular supernatants by label-free mass spectrometry (N = 5). Dark circles indicate adj. p-value < 0.05. In (C) – (E), (H) * denotes immature pro-enzymes.

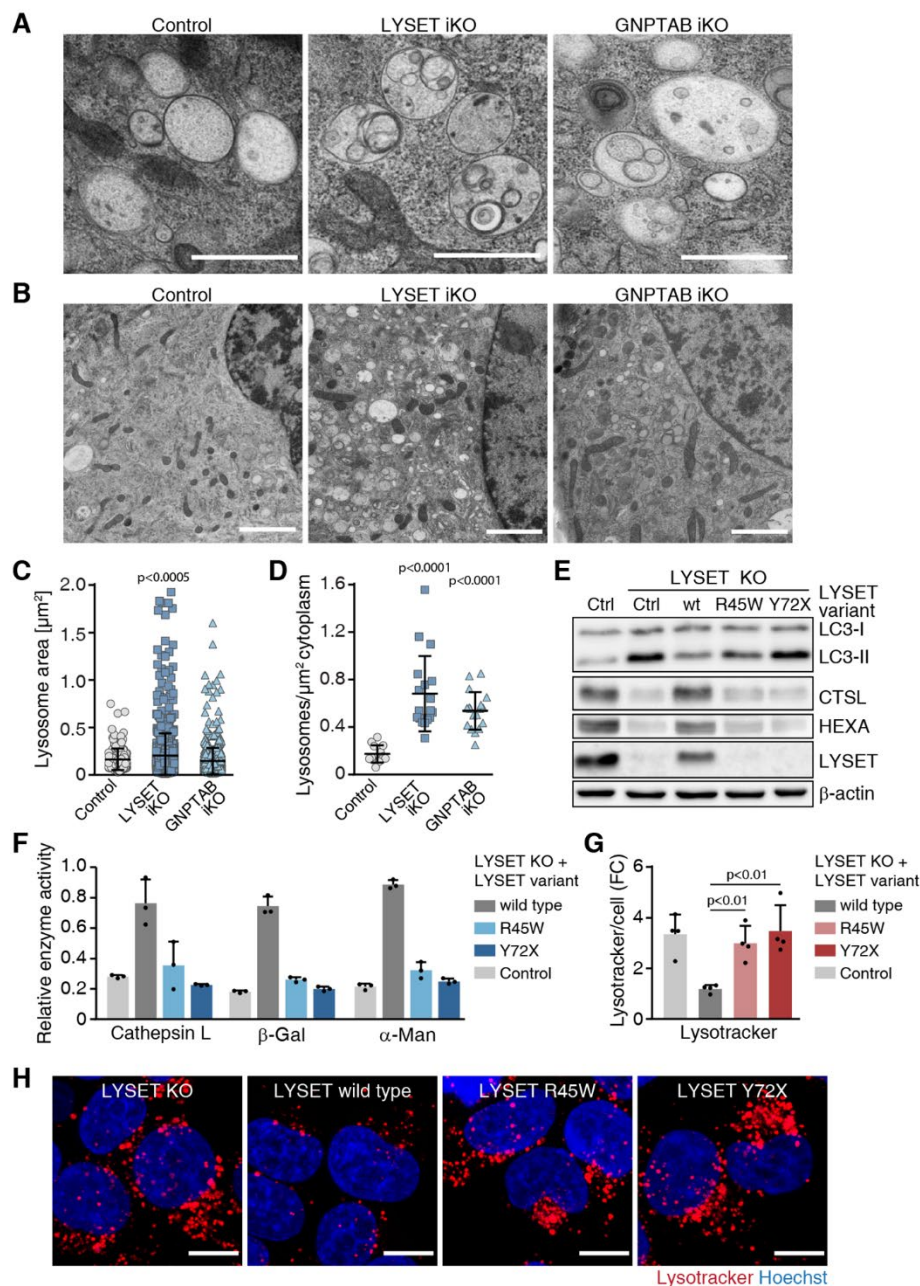


Fig. 4. LYSET deficiency causes lysosomal storage disorder phenotypes. (A and B) Electron micrographs showing (A) undigested lysosomal luminal contents, (B) accumulation of lysosomes in LYSET iKO and GNPTAB iKO MEFs. Scale bars = 1 μm . (C and D) Quantification of (C) lysosomal area, (D) lysosomal number of electron micrographs shown in (B). Data are represented as mean $\pm$ SD (control: 13 cells, 235 lysosomes; LYSET KO, GNPTAB KO: 18 cells, > 900 lysosomes). (E to H) Lysosomal phenotypes in LYSET KO HEK 293T cells expressing LYSET wild type or the pathogenic variants R45W or Y72X: (E) LC3 and lysosomal enzyme levels, analyzed by IB; (F) lysosomal enzyme activities relative to control cells; data are represented as mean $\pm$ SD (N = 3); (G) lysotracker accumulation, quantified by flow cytometry; data are represented as replicate mean $\pm$ SEM (N = 4); (H) lysotracker accumulation, analyzed by microscopy. Scale bars = 10 μm . p-values were calculated by unpaired two-sided *t* test with Welch correction.

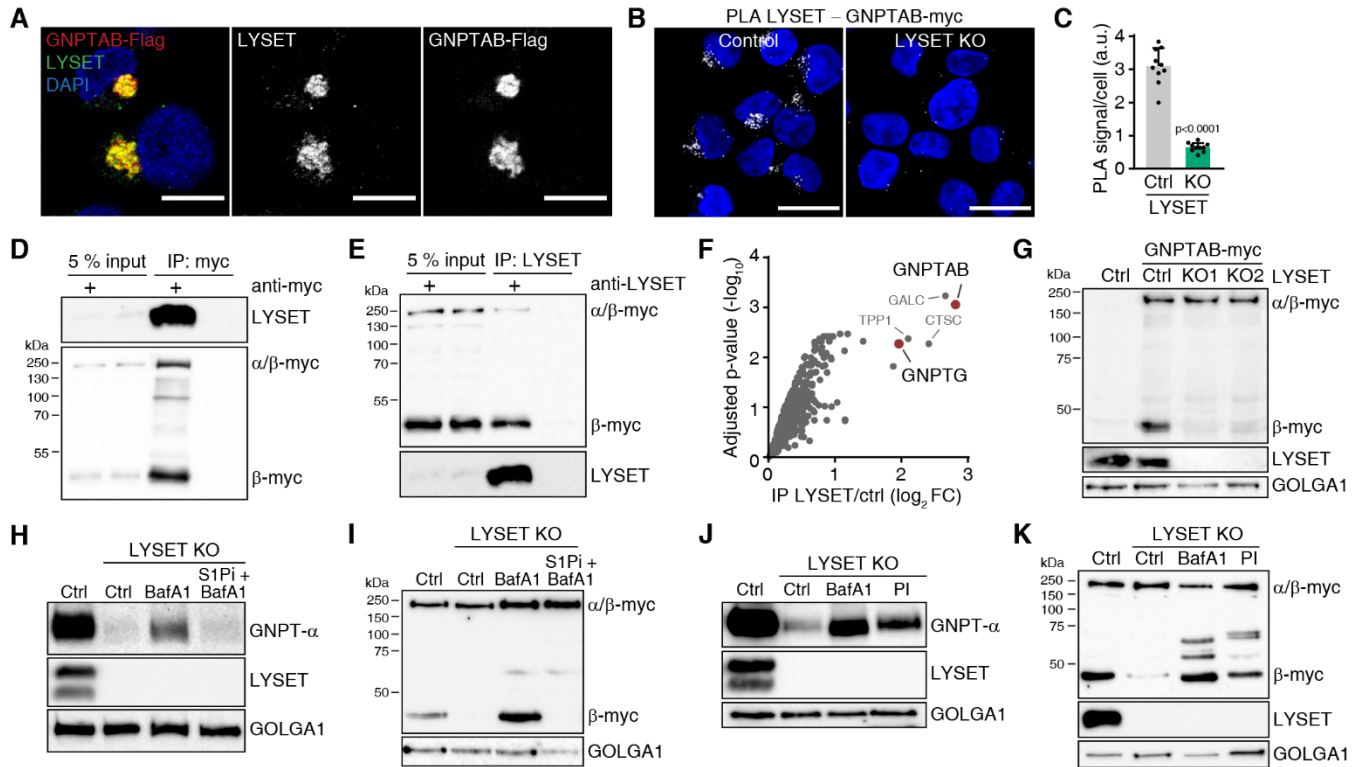


Fig. 5. GlcNAc-1-phosphotransferase is retained in the Golgi apparatus through association with LYSET. (A) Co-localization of LYSET and GNPTAB-Flag in MIA PaCa-2 cells, analyzed by immunofluorescence (IF). Scale bars = 10 μ m. (B) Proximity labeling assay of LYSET and GNPTAB-myc in MIA PaCa-2 cells. Scale bars = 20 μ m. (C) Quantification of PLA signal shown in (B). Data are represented as mean $\pm$ SD (N = 10 fields of view with ≥ 13 cells). p -values were calculated by unpaired two-sided t test with Welch correction. (D and E) Reciprocal Co-IP of transiently expressed LYSET and GNPTAB-myc in HEK 293T cells, analyzed by IB. (F) Endogenous interaction partners of LYSET in MIA PaCa-2 cells, analyzed by SILAC-CoIP mass spectrometry (N = 4). (G) GNPTAB-myc levels in LYSET KO MIA PaCa-2 cells, analyzed by IB. (H and I) Levels of (H) GNPTAB α subunit in SKMEL-30 cells, (I) GNPTAB-myc in MIA PaCa-2 cells deficient for LYSET $\pm$ BafA1 [100 nM] and the S1P inhibitor PF-429242 [5 μ M] for 16 hours, analyzed by IB. (J and K) Levels of (J) GNPTAB α subunit in SKMEL-30 cells, (K) GNPTAB-myc in MIA PaCa-2 cells deficient for LYSET $\pm$ bafilomycin A1 [100 nM] or lysosomal protease inhibitors for 16 hours, analyzed by IB. α/β -myc and β -myc denote the myc-tagged GNPTAB α/β precursor and β subunit, respectively; GNPT- α denotes the α subunit.

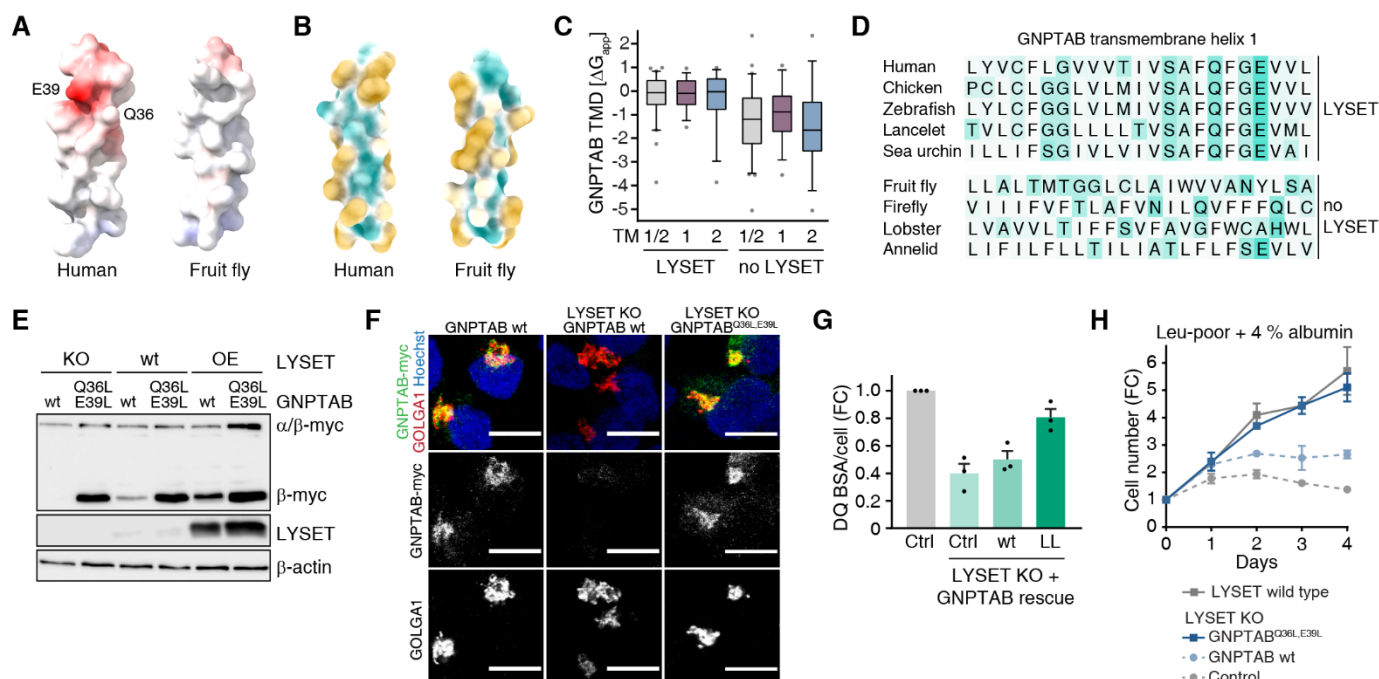


Fig. 6. An unfavorable transmembrane domain renders GlcNAc-1-phosphotransferase dependent on LYSET. (A and B) AlphaFold prediction of transmembrane helix (TM1) of human and fruit fly GNPTAB. (A) negative charge red; positive charge blue; (B) hydrophobic residues ochre; hydrophilic residues turquoise. (C) Prediction of the apparent free energy (ΔG_{app}) for membrane insertion of GNPTAB transmembrane helices 1 and 2 (TM1 and TM2) in animals that possess/lack LYSET homologs. Box plots show 5-95 percentile. A negative ΔG_{app} corresponds to favorable membrane insertion. (D) Sequence comparison of GNPTAB TM1 from animals that possess or lack LYSET homologs. (E) GNPTAB wild type and Q36L,E39L levels in LYSET KO, wild type and LYSET overexpressing (OE) cells, analyzed by IB. α/β -myc and β -myc denote the myc-tagged GNPTAB α/β precursor and β subunit, respectively. (F) GNPTAB wild type and Q36L,E39L localization in LYSET KO cells, analyzed by IF. Scale bars = 10 μ m. (G) DQ BSA degradation in LYSET-deficient cells expressing GNPTAB wild type or Q36L,E39L (LL), quantified by flow cytometry. Data are represented as replicate mean $\pm$ SEM (N = 3). (H) Proliferation of LYSET-deficient cells expressing GNPTAB wild type or Q36L,E39L in leucine-deficient + 4% albumin medium. Data are represented as mean $\pm$ SD (N = 3). Experiments were performed in MIA PaCa-2 cells.

Lysosomal enzyme trafficking factor LYSET enables nutritional usage of extracellular proteins

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